

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

LAB REPORT: CIP-B104 (Computer Forensics Case Study I)

Course Code: CIP-B104

Lab Title: Computer Forensics Case Study & Evidence Integrity Analysis

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User Status: Digital Forensics Internship Trainee (Learning Level)

Dept: Cybersecurity

Program: DFIT (Digital Forensics and Incident Response)

Platform Used: Windows Host / Command Prompt (CertUtil) / Exterro FTK Imager / Text Analysis (Notepad)

1. Executive Summary

This comprehensive digital forensics report documents the step-by-step investigation executed for the CIP-B104 case study. The primary objective of this exercise was to establish strict provenance control through cryptographic hash verification, inspect a raw USB physical disk image (RHINOUSB.dd) using advanced forensic mounting tools, perform unallocated space extraction, and thoroughly analyze historical HTTP proxy/web traffic logs (rhino.log, rhino2.log, rhino3.log). Every analytical phase was executed with meticulous procedural accuracy to ensure chain-of-custody compliance and evidentiary integrity.

2. Methodology & Tools Deployed

To achieve accurate forensic acquisition and examination, specific tools and utilities were utilized across distinct phases of the workflow:

Cryptographic Integrity & Hashing: Windows Native Command Prompt (CMD) utilizing certutil -hashfile to calculate SHA256 checksums manually.

Disk Image Mounting & Carving: Exterro FTK Imager (v4.7.3.81) used for mounting the raw binary image, identifying the file system structure, and exporting unallocated clusters.

Log & Artifact Inspection: Standard text editing environments (Notepad) utilized for inspecting raw MIME types, headers, user agents, and timestamps within the log artifacts.

Documentation & Reporting: Systematic directory segmentation into original, reports, recovered, and screenshots to maintain structural order.

3. Evidence Acquisition & Provenance Control (Task 1)

To guarantee that the evidence files had not been tampered with or altered prior to analysis, manual cryptographic verification was performed via the command line. The generated SHA256 checksums were compiled and stored in hash_manifest.txt within the reports directory:

RHINOUSB.dd (Raw USB Image):

ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65

rhino.log: 64e6d55b76660eb3aaa41572c1d04e4452510a343bbf42e844424827dedfddb2

rhino2.log: 41939d5de0556b70279056572dee44b6fd84cd05b0788cfd0b6f52f37b161dde

rhino3.log: 7b0304f5e88a30c305a99b5a1e2977bcd5b9c94da6f458f887bb39966bbfc46

4. File System & Unallocated Space Carving (Task 2)

File System Identification: Upon loading RHINOUSB.dd into Exterro FTK Imager, the evidence structure mapped successfully as a FAT16 file system.

Unallocated Space Export: By drilling down into the [unallocated space] node of the evidence tree, specific block segments (00004, 25604, and 51204) containing deleted or fragmental artifacts were identified. These blocks were exported and isolated inside the designated recovered folder for subsequent carving procedures.

5. Network Log & Traffic Artifact Analysis (Task 3)

The examination of the log files (rhino.log, rhino2.log, and rhino3.log) provided clear visibility into historical web interactions:

Browser & Environment Fingerprinting: The traffic traces expose legacy browser requests running under Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; .NET CLR 1.1.4322).

Timeline Verification: Consistent HTTP transactions directed towards www.google.com across April 2004 (specifically Mon, 26 Apr 2004 and Wed, 28 Apr 2004) establish an accurate historical timeline framework.

Payload Metrics: Standard HTML response parameters, including a Content-length: 2931 structure and inline image references, confirm active web session logging.

6. Conclusion

The forensic investigation for CIP-B104 has been successfully executed from end to end. Through rigorous command-line hash extraction, precise FTK Imager parsing, and structural log correlation, all artifacts have been properly accounted for, hashed, and categorized.